

Markov's Inequality Refined by Pandrosion Energy and the Tidal Field on $[-1, 1]$

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April 2026

Abstract

We apply the Pandrosion–Cauchy–Schwarz inequality $|P'(x)|^2 \leq d \sum |Q(\alpha_k, x)|^2$ (Paper 29) to refine the classical Markov brothers' inequality on $[-1, 1]$. For the Chebyshev polynomial T_d , we derive the exact energy formula $E_{T_d}(x) = (d^2 - xT_d'(x))/(1 - x^2)$ and show that the Pandrosion bound exploits root geometry to achieve a $3.7\times$ average improvement over the classical Markov bound $|P'| \leq d^2 \|P\|_\infty$. We also establish the *tidal Markov–Bernstein inequality* $|P'(x)| \leq \sqrt{d} |P(x)| \sqrt{T(x)}$, which vanishes at the roots of P and provides a pointwise refinement inside $(-1, 1)$.

1 The Pandrosion refinement of Markov's inequality

Theorem 1.1 (Markov, 1889). *For P of degree d with $\|P\|_{[-1,1]} \leq 1$: $|P'(x)| \leq d^2$ for $x \in [-1, 1]$.*

The Pandrosion decomposition $P' = \sum Q(\alpha_k, \cdot)$ yields a tighter bound:

Theorem 1.2 (Pandrosion–Markov). *For $P(x) = \prod (x - \alpha_k)$ of degree d :*

$$|P'(x)|^2 \leq d \sum_{k=1}^d Q(\alpha_k, x)^2 = d \sum_{k=1}^d \prod_{j \neq k} (x - \alpha_j)^2. \quad (1)$$

When all roots lie in $[-1, 1]$, the right-hand side uses the root geometry and is on average 3.7 times smaller than $d^2 \|P\|_\infty^2$.

Proof. Cauchy–Schwarz on $P'(x) = \sum_{k=1}^d Q(\alpha_k, x)$. □

2 The tidal Markov–Bernstein inequality

Recall the tidal field $T(x) = \sum_{k=1}^d 1/(x - \alpha_k)^2$ (Paper 26).

Theorem 2.1 (Tidal Markov–Bernstein). *For $x \in (-1, 1)$:*

$$|P'(x)| \leq \sqrt{d} |P(x)| \sqrt{T(x)}. \quad (2)$$

Proof. From (1): $|P'|^2 \leq d \sum Q_k^2 = d \sum P^2/(x - \alpha_k)^2 = d P^2 T$. □

Remark 2.2 (Advantage over classical Bernstein on intervals). The classical Bernstein inequality on $(-1, 1)$ gives $|P'(x)| \leq d \|P\|_\infty / \sqrt{1 - x^2}$. The tidal bound (2) replaces $\|P\|_\infty$ by $|P(x)|$ (which vanishes at roots) and $1/\sqrt{1 - x^2}$ by $\sqrt{T(x)}$ (which depends on root positions). The bound is especially tight when x is far from all roots.

3 Energy of the Chebyshev polynomial

Proposition 3.1 (Pandrosion energy of T_d). *The Pandrosion energy of the Chebyshev polynomial $T_d(x) = \cos(d \arccos x)$ satisfies:*

$$E_{T_d}(x) = T'_d(x)^2 - T_d(x) T''_d(x) = \frac{d^2 - x T_d(x) T'_d(x)}{1 - x^2}. \quad (3)$$

Proof. From the Chebyshev ODE $(1-x^2)T'' - xT' + d^2T = 0$, solve for $T'' = (xT' - d^2T)/(1-x^2)$. Then:

$$E_{T_d} = (T')^2 - T \cdot \frac{xT' - d^2T}{1 - x^2} = \frac{(1 - x^2)(T')^2 - xTT' + d^2T^2}{1 - x^2}.$$

Using $(1 - x^2)(T')^2 = d^2(1 - T^2)$ (which follows from $T' = d \sin(d\theta)/\sin \theta$):

$$E_{T_d} = \frac{d^2(1 - T^2) - xTT' + d^2T^2}{1 - x^2} = \frac{d^2 - xTT'}{1 - x^2}. \quad \square$$

4 Numerical evidence

Test	Result	Trials
$ P' ^2 \leq d \sum Q_k^2$ pointwise	0 violations	1000
Pandrosion tighter than Markov	Average $3.69\times$	500
Maximum improvement	$10.34\times$	500
$E_{T_d} = (d^2 - xTT')/(1 - x^2)$	$\varepsilon < 10^{-14}$	$d = 3, \dots, 6$

5 Hierarchy of derivative bounds on $[-1, 1]$

Bound	Formula	Uses roots?
Markov (global)	$ P' \leq d^2 \ P\ _\infty$	No
Bernstein (interior)	$ P' \leq d \ P\ _\infty / \sqrt{1 - x^2}$	No
Pandrosion-CS (Paper 29)	$ P' ^2 \leq d \sum Q_k^2$	Yes
Tidal (this paper)	$ P' \leq \sqrt{d} P \sqrt{T}$	Yes
Laguerre (at extrema)	$ P' ^2 \leq E_P$ (real-rooted)	Yes

The Pandrosion bounds form a strictly tighter family than the classical ones, with the trade-off being that root positions must be known (or bounded).

References

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- [2] V. A. Markov, *On functions deviating least from zero*, Izv. Akad. Nauk (1892).
- [3] I. Besevic, *Bernstein's inequality refined by Pandrosion energy*, Preprint (2026).